Category-Level Assignment for Cross-Domain Semantic Segmentation in Remote Sensing Images

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***Abstract*— Deep learning-based semantic segmentation has made great progress in understanding very-high-resolution (VHR) remote sensing images (RSIs). However, large-scale applications are still limited. The main reason is that diverse imaging modes and geographical differences make it difficult to transfer a model trained in the source domain to the target domain. To solve this problem, unsupervised domain adaptation (UDA) for VHR RSIs has received some attention, but the accuracy of cross-domain semantic segmentation still needs to be improved. Currently, one reasonable proposal for improving accuracy is to take a close look at the category-level information. In this article, we reveal an integer programming mechanism for modeling the category-level relationship between the source and target domains. The mechanism is based on the solution of the assignment problem, and thus, the proposed method is called category-level assignment for UDA (ClA-UDA). In ClA-UDA, a category-level assignment problem with additional constraints is defined for UDA tasks, and the solution is provided. Based on the solution, an assignment-based image-to-image transferring algorithm (AIT) is first proposed to transfer the source-domain images based on the style of the target-domain images. AIT minimizes a weighted discrepancy and provides an analytical solution for the transfer. Two assignment-based alignment losses are then introduced to align the source and target domains based on the category-level relationship in a concise way. To validate the performance of ClA-UDA, three VHR RSI datasets are employed, and six UDA tasks are designed. Extensive experiments are conducted, and the results demonstrate the superiority of ClA-UDA compared with the existing methods.**

***Index Terms*— Assignment problem, category-level relationship, semantic segmentation, unsupervised domain adaptation (UDA).**

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I. Introduction

SEMANTIC segmentation for very-high-resolution (VHR) remote sensing images (RSIs) assigns a label to each pixel, playing a vital role in understanding large-scale ground information [1]. Benefiting from the fully convolutional network (FCN) [2] and FCN-based networks [3], [4], the accuracy of the semantic segmentation of VHR RSIs has been considerably improved. However, the requirement for the performance of a semantic segmentation network to be guaranteed is that the training samples (in the source domain) and the testing data (in the target domain) are strictly drawn from the same distribution [5]. This premise limits the applications of semantic segmentation networks. Specifically, for a large-scale remote sensing task, VHR RSIs contain diverse geographic environments and may be acquired by different imaging sensors. It is difficult to ensure that all potential geographic environments and multisource VHR RSIs are included in the training samples [6]. When the training samples do not contain the same data form and geographic environment as the target domain, different data distributions appear. These distribution differences are often known as domain shifts [5].

A common solution to alleviate the influence of domain shifts is unsupervised domain adaptation (UDA) [7]. UDA adapts a source model to the target domain using unlabeled target data [8]. Currently, most methods can be divided into two main groups: alignment-based [9], [10] and self-training-based [8], [11]. In alignment-based methods, generative adversarial nets (GANs) [12] are commonly used to align the marginal distributions [13]. The performance of these methods is validated using natural image datasets. However, these methods cannot fully address the domain shifts in VHR RSIs, mainly because VHR RSIs have their own distinct characteristics, such as imaging mode differences, spatial resolution variations, and spatial layout changes [5]. To alleviate domain shifts in VHR RSIs, the latest studies [5], [6] often combine alignment-based and self-training-based strategies. In this case, a retraining or progressive training process is required. With this combination, the reported mean intersection-over-union (mIoU) score from Potsdam to Vaihingen in [5] exceeds 50% when all six classes are considered. However, the training difficulty hinders the application of these methods.

Meanwhile, the category-level adversarial network (CLAN) [13] and fine-grained adversarial domain adaptation (FADA) [14] model category-level information to aid UDA. For VHR

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RSIs, [15] and [16] further enhance this category-level modeling strategy and alleviate the training difficulty. However, it is difficult to give a mathematical or intuitive explanation of these methods, and the accuracy remains difficult to improve. The class-aware GAN (CaGAN) [17] presents an intuitive way to model category-level information. However, the steps of the CaGAN are complex, and an alternating and asymptotic training scheme [11], which is more complex than the self-training strategy, is required to progressively train the network. Therefore, the question arises whether there is a mathematical method that can effectively model the category-level relationship between the source and target domains. Based on this category-level relationship, a higher accuracy than that obtained by existing methods could be obtained without retraining.

In this article, the category-level assignment for UDA (ClA-UDA) is proposed to meet this objective. The code of ClA-UDA will be available at: https://github.com/yu-ni1989/ClA-UDA. In ClA-UDA, the assignment problem, which is a specific task in integer programming [18], is employed to model the category-level relationship between the source and target domains. Combining the characteristics of UDA tasks, we define a category-level assignment problem with additional constraints and provide the solution. Based on the solution, a novel assignment-based image-to-image transferring algorithm (AIT) and two assignment-based alignment losses are proposed. The main contributions are summarized as follows.

1) We reveal a crucial mechanism for modeling the category-level relationship between the source and target domains. The mechanism defines a category-level assignment problem with additional constraints and provides the solution. This will be pioneering work in considering category-level alignment from the perspective of integer programming for UDA tasks.
2) The AIT is proposed to transfer source-domain images into target-domain styles by minimizing weighted discrepancy. AIT not only provides an analytical solution to image style transferring but also takes the category-level relationship into consideration. AIT plays a vital role in accuracy improvement.
3) Two assignment-based alignment losses are developed using the category-level relationship. These losses align the source and target domains in a concise way. Using only a one-stop training process, ClA-UDA can produce higher accuracy than existing methods.

II. Related Work

As suggested above, existing UDA methods can be roughly divided into two main groups: alignment-based and self-training-based. Here, to review the related work in accordance with the properties of ClA-UDA, we add a group of UDA methods for RSIs. Furthermore, the methods that model the category-level information are detailed in each group.

A. Alignment-Based Methods

Alignment-based methods align the source and target domains in image, feature, and output spaces. The methods that align domains in image space benefit from image-to-image transferring studies, such as PixelDA [19], CycleGAN [20], DiscoGAN [21], and DualGAN [22]. These methods [23], [24] first adapt source-domain images to appear as if drawn from the target domain. The model is then trained based on the adapted source-domain images to meet the UDA objective. Unlike the methods that align in image space, those that align in feature space force the alignment of marginal distributions in feature space through adversarial learning. Researchers have proposed various adversarial losses to improve the performance of adversarial learning for feature-level alignment, as in [25] and [26]. Meanwhile, Tsai et al. [27] validated that adversarial learning in output space is more powerful. A range of methods has employed alignment in output space. For example, maximum classifier discrepancy (MCD) [28] maximized the discrepancy between two task-specific classifiers adversarially in output space. Subsequently, Lee et al. [29] introduced the sliced Wasserstein distance [30] into the MCD framework.

To fully use the category-level information, CLAN [13] integrated the class-aware information into the adversarial learning in output space and aligned each class with an adaptive adversarial loss. Subsequently, FADA [14] adopted a fine-grained domain discriminator that not only acts as a domain distinguisher but also differentiates the domains at the category level. Wang et al. [10] proposed a differential treatment method that divides existing categories into "stuff" and "things." Different processing strategies were designed for the stuff and thing categories.

B. Self-Training-Based Methods

Self-training methods usually involve three steps. First, a segmentation model is trained using source-domain ground truth. The pseudolabels for the target domain are then generated based on the trained model. Finally, the model is fine-tuned using a retraining procedure using the pseudolabels. This idea has been employed to further enhance the adaptation performance of alignment-based methods, as in [10]. However, the commonly used self-training steps have instinctual drawbacks. To avoid the gradual dominance of large classes in pseudolabel generation, Zou et al. [7] proposed a class-balanced self-training (CBST) framework that introduces spatial priors to refine the generated labels. Later, scholars discovered that the pseudolabels were noisy and the target features were dispersed due to the discrepancy between the source and target domains. A range of studies [8], [31] then focused on denoising during pseudolabel generation. Specifically, ProDA [31] online corrected the soft pseudolabels according to the relative feature distances to prototypes. In particular, the prototypes were updated on the fly. The progressive feature alignment network (PFAN) [11] exploited the intraclass variation in the target domain. Furthermore, PFAN developed an easy-to-hard transfer strategy and a progressive training process. The easy-to-hard strategy and the progressive training process have been continuously improved in subsequent studies [5], [6].

These alignment-based and self-training-based methods achieved good results in natural image sets. However, they

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could not optimally address the domain shifts in RSIs because of the distinct characteristics of RSIs [5].

C. UDA Methods for RSIs

To deal with the complex domain shifts in RSIs, various studies specific to UDA for RSIs have been conducted. Initially, GANs were employed in [15] and [32], and conditional GANs [33] were employed in [34]. However, the accuracy was not effectively improved by their experiments. To this end, the GAN procedure was further enhanced in [35]. Specifically, a novel domain similarity discriminator was proposed to jointly consider both domains. Later, a full-space domain adaptation method that aligns the source and target domains in image, feature, and output spaces based on GANs was proposed in [36]. Meanwhile, based on CycleGAN [20], the multisource problem in RSIs was considered in [37] to achieve joint domain adaptation in an end-to-end manner. However, the accuracy was not effectively improved.

Recently, the steps of UDA methods for RSIs have been made more complicated to further improve their accuracy. For example, [1] integrated multitask learning into an entropy adversarial learning framework, [38] learned a deep semantic segmentation network under multiple weakly supervised constraints, and BiFDANet [39] obtained better UDA performance by optimizing the semantic segmentation network in two opposite directions. Specifically, the curriculum-style cross-domain adaptation (CCDA) [5] used a two-stage procedure that aligns domains in an easy-to-hard way and achieved higher accuracy. In CCDA, a retraining process is required to deal with the hard-to-adapt target patches. Meanwhile, semantic- and entropy-level classifiers were proposed to model both the local semantic and global spatial layout information. Compared to [5], some methods, such as the cross teacher–student network (CTSN) [40], the stepwise domain adaptive segmentation network with covariate shift alleviation (Cov-DA) [6], iterative intradomain adaptation (IterDANet) [41], and CaGAN [17], require more training stages and have more complicated architecture.

Among these studies, CaGAN [17] modeled category-level information. To address the issue that the same class of objects may have different spectra while different classes of objects may have the same spectra, CaGAN modeled the intraclass and interclass discrepancies in an intuitive way. However, this issue exists not only in UDA tasks but also in fully supervised semantic segmentation tasks. These tasks obtained good results using an FCN-based network, just like the baseline model in UDA architectures. Furthermore, the label information of the target domain is inaccurate, which challenges the CaGAN mechanism. In this case, alternating and iterative training [11] was required to obtain reliable cross-domain segmentation results. Unlike CaGAN, the attention mechanism was employed by [15] and [16] to model category-level information. Chen et al. [15] proposed category certainty attention to find the misaligned regions at the category level. A classwise domain predictor was then proposed to produce a category-level adversarial loss. Chen et al. [16] first eliminated the interference between different classes via a novel class-clear module. Entropy-based regional attention was then proposed to align the regions that were difficult to transfer. However, [15] and [16] are difficult to reasonably explain mathematically or intuitively.

III. Methods

To better illustrate our method, we first introduce the overall architecture of ClA-UDA. The key innovations, including category-level assignment, assignment-based image-to-image transferring (AIT), and assignment-based alignment losses, are then presented.

A. Overall Architecture of ClA-UDA

ClA-UDA only needs to be used during training to learn the parameters of the semantic segmentation backbone. After the training process, the semantic segmentation backbone can directly obtain the result of the target domain. Since the semantic segmentation backbone network (such as DeepLabV3 [42]) is well known to readers and is not the research focus of this article, we do not present the details. Fig. 1 is the overall architecture of ClA-UDA. The feature f^s and classifier output O^s of the source domain and the feature f^t and classifier output O^t of the target domain are generated by the same semantic segmentation backbone with shared weights.

As shown in Fig. 1, the inputs of ClA-UDA are the randomly selected source-domain image patch I^s and target-domain image patch I^t. We know the ground-truth G^s of I^s and the ground-truth of I^t are absent. First, the target-domain feature f^t and classifier output O^t are generated based on I^t. Using category-level assignment, the category-level relationship between O^t and G^s is generated. The relationship is formed by the solution $\hat{X}^g$ of the assignment problem and the weight matrix W^m. Next, using $\hat{X}^g$ and W^m, AIT transforms I^s into a target-like image I^{st} that covers the same area as I^s and is stylistically similar to I^t. The source-domain feature f^s and classifier output O^s are then generated based on I^{st}. Using category-level assignment, the category-level relationship $\hat{X}^o$ between O^t and O^s is generated. At this time, all the variables for computing losses are obtained. As shown in Fig. 1, a supervision loss and two assignment-based alignment losses are computed. Finally, these losses are summed and backpropagated.

B. Category-Level Assignment

The areas covered by the source- and target-domain image patches (I^s and I^t) are different. When we transfer the image style between I^s and I^t and align the source and target domains, we should force the spectra of source- and target-domain pixels belonging to the same category closer. To meet this objective, the category-level relationships between the source and target domains are necessary. O^s, G^s, and O^t carry the category-level information of the source and target domains and lay the data foundation for modeling the category-level relationships. In this article, we define a novel category-level assignment problem with additional constraints. Based on this, the category-level relationships between O^t and G^s and between O^t and O^s are generated. In the following, to facilitate the understanding of this section, the standard assignment problem is first introduced.

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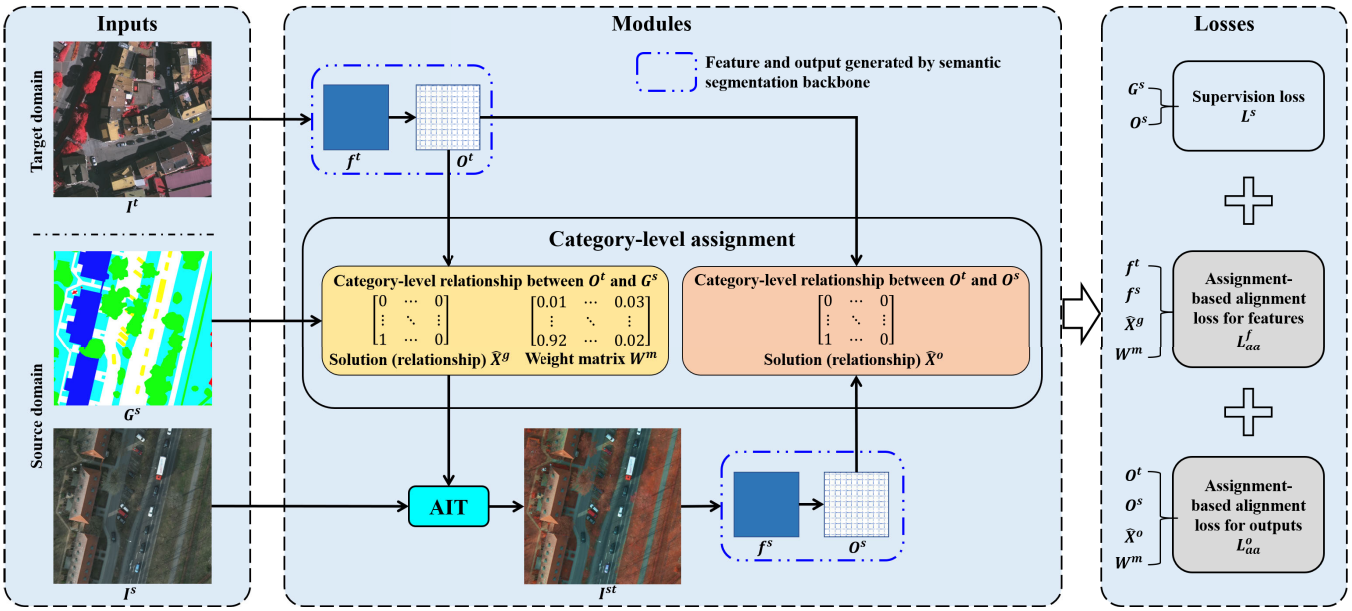


Fig. 1. Overall architecture of ClA-UDA. The inputs of ClA-UDA are the randomly selected source-domain image patch I^s with ground-truth G^s, and the target-domain image patch I^t. The category-level assignment and AIT are the key modules of ClA-UDA. Among the losses, assignment-based alignment losses L^f_{aa} and L^o_{aa} are used to align the source and target domains.

1) Standard Assignment Problem: Let us consider n workers and n jobs, with the cost of each worker for each job. The standard assignment problem is to assign each worker to only one job and vice versa so that the given measure of the overall cost is optimized. In general, we use a cost matrix $\mathcal{C} = [c_{i,j}, i = 1, \ldots, n; j = 1, \ldots, n]$ to record the cost of each worker for each job as follows:

$$\begin{bmatrix} c_{1,1} & \cdots & c_{1,j} & \cdots & c_{1,n} \\ \vdots & \ddots & & & \vdots \\ c_{i,1} & & c_{i,j} & & c_{i,n} \\ \vdots & & & \ddots & \vdots \\ c_{n,1} & \cdots & c_{n,j} & \cdots & c_{n,n} \end{bmatrix} \tag{1}$$

where $c_{i,j}$ is the cost when worker i is assigned to job j. Suppose that $x_{i,j}$ represents the assignment of worker i to job j such that

$$x_{i,j} = \begin{cases} 1, & \text{if worker } i \text{ is assigned to job } j \\ 0, & \text{otherwise} \end{cases} \tag{2}$$

then the mathematical model of the assignment problem can be stated as

$$\min Z = \sum_{i=1}^{n} \sum_{j=1}^{n} c_{i,j} x_{i,j} \tag{3}$$

subject to the following constraints:

$$\begin{cases} \sum_{i=1}^{n} x_{i,j} = 1, & i = 1, \ldots, n \\ \sum_{j=1}^{n} x_{i,j} = 1, & j = 1, \ldots, n \\ x_{i,j} = 0 \text{ or } 1, & i = 1, \ldots, n; \ j = 1, \ldots, n. \end{cases} \tag{4}$$

This model can be solved by a range of methods, such as those in [43] and [44]. In this article, we used the method in [44] as the default. The optimal solution of the model can be denoted by a matrix $X^* = [x^*_{i,j}, i = 1, \ldots, n; j = 1, \ldots, n]$. X^* only contains n elements that are equal to 1. This means that X^* forms n assigned pairs. For example, if there are five workers and five jobs, X^* is

$$\begin{bmatrix} 0 & 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 \\ 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 \end{bmatrix}. \tag{5}$$

In Formula (5), $x^*_{3,5} = 1$ indicates an assigned pair $\langle 3, 5 \rangle$, which means that worker 3 is assigned to job 5. The assumptions of the standard assignment problem are fairly restrictive. Many nonstandard assignment problems do not satisfy these assumptions. However, there are many methods, such as adding dummy workers, cloning jobs, and setting large values in $\mathcal{C}$, through which nonstandard assignment problems can be transformed into the standard form so that the methods for standard assignment problems can be used. To model the relationship between the source and target domains, each source-domain pixel is considered a worker, and each target-domain pixel is considered a job.

2) Definition and Solution of Category-Level Assignment: In addition to the objective function and constraints in (3) and (4), there are two additional constraints in the category-level assignment problem.

a) Additional Constraint A: If a source-domain pixel p^s_i and a target-domain pixel p^t_j belong to different categories, p^s_i cannot be assigned to p^t_j.

b) Additional Constraint B: To model the category-level relationship between source and target domains as comprehensively as possible, as many pixels as possible should be assigned.

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To finish the definition of a category-level assignment, we formulate cost matrix $\mathcal{C}$, which is computed using cosine similarity. Suppose that the category-level assignment problem is defined in a space of $\mathcal{R}^{N^c \times N^{\text{cla}}}$, where N^c is the number of existing categories and N^{cla} is $H^{\text{cla}} \times W^{\text{cla}}$. The cosine similarity between the ith pixel p_i^s in the source domain and the jth pixel p_j^t in the target domain is computed as follows:

$$\text{Cos}_{i,j} = \frac{p_i^s \cdot p_j^t}{\|p_i^s\| \|p_j^t\|} \tag{6}$$

where $i = 1, \ldots, N^{\text{cla}}$ and $j = 1, \ldots, N^{\text{cla}}$. The maximum value Cos^{Max} of these cosine similarities is determined. Then, each element $c_{i,j}$ in $\mathcal{C}$ is computed as follows:

$$c_{i,j} = \text{Cos}^{\text{Max}} - \text{Cos}_{i,j}. \tag{7}$$

Using the defined $\mathcal{C}$, the assignment programming finds the closest (or most similar) pixel in the source domain for each pixel in the target domain.

Additional Constraint A means that some workers cannot be assigned to certain jobs. Hillier and Lieberman [45] gives a method that sets a value of M much larger than the costs for corresponding positions in $\mathcal{C}$. As a result, each element $\hat{c}_{i,j}$ in the modified cost matrix $\hat{\mathcal{C}}$ is

$$\hat{c}_{i,j} = \begin{cases} M, & \text{if worker } i \text{ cannot be assigned to job } j \\ c_{i,j}, & \text{otherwise.} \end{cases} \tag{8}$$

Then, the solution that minimizes Z in (3) can be computed by [43] or [44] using $\hat{\mathcal{C}}$. This method is usually used when the number of M is small. In this case, there is no element with a cost of M in the solution. However, the UDA task for semantic segmentation has different characteristics.

In each training iteration for UDA, the source-domain image patch I^s and the target-domain image patch I^t are randomly selected. I^s may contain categories that do not exist in I^t, and vice versa. Furthermore, the number of pixels belonging to a category in I^s is not equal to that in I^t. To satisfy Additional Constraint A, a large range of costs in $\hat{\mathcal{C}}$ are set to M, as shown in (8). Under the constraints in (4), the solution $X^* = [x_{i,j}^*, i = 1, \ldots, n; j = 1, \ldots, n]$ that minimizes Z in (3) obtained by [43] or [44] always includes a range of elements whose corresponding costs are equal to M. However, X^* can be the initial solution to the category-level assignment problem.

Each element $\hat{x}_{i,j}^*$ in the final solution $\hat{X}^*$ is obtained as follows:

$$\hat{x}_{i,j}^* = \begin{cases} 0, & \text{if } \hat{c}_{i,j} = M \\ x_{i,j}^*, & \text{otherwise.} \end{cases} \tag{9}$$

Obviously, $\hat{X}^*$ satisfies Additional Constraint A. At the same time, $\hat{X}^*$ can satisfy Additional Constraint B as long as M is large enough. As mentioned before, the category-level assignment problem is defined in $\mathcal{R}^{N^c \times N^{\text{cla}}}$. This means that both the source and target domains have N^{cla} pixels for assignment. Here, we set M to $N^{\text{cla}} + 1$. The proof that this setting of M guarantees that $\hat{X}^*$ satisfies Additional Constraint B is shown in Appendix A.

3) Category-Level Relationship Between O^t and G^s: To make category-level assignments between $O^t \in \mathcal{R}^{N^c \times H^o \times W^o}$ and $G^s \in \mathcal{R}^{1 \times H \times W}$, we should transform O^t and G^s into $\mathcal{R}^{N^c \times N^{\text{cla}}}$, where the category-level assignment problem is defined.

First, we have to convert G^s into the one-hot form $\bar{G}^s$. Then, the average pooling operator $\text{APool}_{\{(\text{kernel size}),(\text{stride}),(\text{padding})\}}$ is performed to downsample O^t and $\bar{G}^s$ into $\mathcal{R}^{N^c \times H^{\text{cla}} \times W^{\text{cla}}}$ as follows:

$$\begin{cases} \hat{G}^s = \text{APool}_{\{(2\tau_g^h+1, 2\tau_g^w+1), (\tau_g^h, \tau_g^w), (\tau_g^h, \tau_g^w)\}}(\bar{G}^s) \\ \hat{O}^t = \text{APool}_{\{(2\tau_o^h+1, 2\tau_o^w+1), (\tau_o^h, \tau_o^w), (\tau_o^h, \tau_o^w)\}}(O^t) \end{cases} \tag{10}$$

where $\tau_g^h = H//H^{\text{cla}}$, $\tau_g^w = W//W^{\text{cla}}$, $\tau_o^h = H^o//H^{\text{cla}}$, and $\tau_o^w = W^o//W^{\text{cla}}$. $//$ signifies division with an integral result (that discards the remainder). Average pooling operators are important because they enhance the interactions between different classes of objects in complex scenes, especially for $\bar{G}^s$. Since N^{cla} is $H^{\text{cla}} \times W^{\text{cla}}$, $\hat{G}^s$, and $\hat{O}^t$ can be reshaped into $\mathcal{R}^{N^c \times N^{\text{cla}}}$.

Each element $c_{i,j}^g$ in the cost matrix $\mathcal{C}^g$ is computed based on $\hat{G}^s$ and $\hat{O}^t$ using (6) and (7). Suppose p_i^s and p_j^t are the ith and jth pixels in $\hat{G}^s$ and $\hat{O}^t$. The label l_i^s of p_i^s and the label l_j^t of p_j^t are determined as follows:

$$\begin{cases} l_i^s = \arg\max_{k \in [1,\ldots,N^c]} (\hat{G}^s[:, i]) \\ l_j^t = \arg\max_{k \in [1,\ldots,N^c]} (\hat{O}^t[:, j]) \end{cases} \tag{11}$$

where $\hat{G}^s[:, i] \in \mathcal{R}^{N^c \times 1}$ is the ith column of $\hat{G}^s$, and $\hat{O}^t[:, j]$ has a similar meaning. $\arg\max_{k \in [1,\ldots,N^c]}$ finds the position where the maximum value is located.

Subsequently, the modified cost matrix $\hat{\mathcal{C}}^g = [\hat{c}_{i,j}^g, i = 1, \ldots, n; j = 1, \ldots, n]$ is obtained using a process similar to that in (8)

$$\hat{c}_{i,j}^g = \begin{cases} M, & \text{if } l_i^s \neq l_j^t \\ c_{i,j}^g, & \text{otherwise.} \end{cases} \tag{12}$$

Finally, the solution $\hat{X}^g$ can be found using the method presented in Section III-B2 and $\hat{\mathcal{C}}^g$. As shown in Formula (5), $\hat{X}^g$ forms the category-level relationship between O^t and G^s.

To further enhance the reliability of the category-level assignment, a weight matrix $W^m = [w_{i,j}^m, i = 1, \ldots, n; j = 1, \ldots, n]$ is defined. The cosine similarity can naturally represent the reliability of the assigned pixel pair $\langle i, j \rangle$. $w_{i,j}^m$ is computed as in (6). Moreover, G^s shows the source-domain ground truth. A larger $w_{i,j}^m$ also indicates higher confidence in the prediction of the jth pixel in O^t. This provides a convenient way to relieve the negative effects of noise in O^t. Therefore, W^m is employed in both the AIT and assignment-based alignment losses.

4) Category-Level Relationship Between O^t and O^s: O^t and O^s are first transformed into $\mathcal{R}^{N^c \times N^{\text{cla}}}$ using a process similar to that used in Section III-B3. Then, the cost matrix $\hat{\mathcal{C}}^o$ is computed using (6), (7), and (12). The programming process is similar to that of Section III-B3. We use $\hat{X}^o$ to denote the solution that forms the category-level relationship between O^t and O^s.

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C. Assignment-Based Image-to-Image Transferring

AIT is an unsupervised image-to-image transferring algorithm. Given a source-domain image patch I^s and a target-domain image patch I^t, AIT converts I^s to a target-like image I^{st}. I^{st} has an image style similar to I^t and covers the same area as I^s. We assume that I^{st} can be produced through a linear transformation. Mathematically

$$I^{st} = (I^s - \mu^s)T^m + \mu^t \tag{13}$$

where μ^s and μ^t are the mean vectors of I^s and I^t, respectively. The key to AIT is the estimation of the transferring matrix T^m.

We transform the estimation of T^m as a weighted discrepancy minimization process that forces the spectra of source- and target-domain pixels in the same category closer. To meet this objective, the category-level relationship between O^t and G^s formed by $\hat{X}^g$ and W^m is employed. For easy retrieval, $\hat{X}^g$ is transformed into two vectors: $\hat{x}_s^v$ and $\hat{x}_t^v$. $\hat{x}_s^v$ presents the pixel index of the source domain, and $\hat{x}_t^v$ presents the pixel index of the target domain. The $\hat{x}_s^v(k)$th pixel in the source domain is assigned to the $\hat{x}_t^v(k)$th pixel in the target domain, where $\hat{x}_s^v(k)$ is the kth element of $\hat{x}_s^v$ and $\hat{x}_t^v(k)$ is the kth element of $\hat{x}_t^v$. To determine the pixel correspondence between the image patches (I^s and I^t) and $\hat{X}^g$, we averagely pool I^s and I^t so that the resolution of them is the same as the resolution of the category-level assignment space $\mathcal{R}^{N^c \times H^{\text{cla}} \times W^{\text{cla}}}$, where $H^{\text{cla}} \times W^{\text{cla}}$ is equal to N^{cla}. To meet the requirements of (13), we then transform I^s and I^t as follows:

$$\begin{cases} I^s = I^s - \mu^s \\ I^t = I^t - \mu^t. \end{cases} \tag{14}$$

Suppose I^s and I^t have C^s and C^t channels, respectively, and T^m is composed of C^s rows and C^t columns. To minimize the discrepancy, the objective function is formulated as follows:

$$\Psi = \sum_{k=1}^{N^x} w^m_{\hat{x}_s^v(k),\hat{x}_t^v(k)} \sum_{l=1}^{C^t} D_k(l)^2 \tag{15}$$

where N^x is the number of assigned pixel pairs (the length of $\hat{x}_s^v$ or $\hat{x}_t^v$). $w^m_{\hat{x}_s^v(k),\hat{x}_t^v(k)}$ is the weight for the kth pair of assigned pixels $p^s_{\hat{x}_s^v(k)}$ and $p^t_{\hat{x}_t^v(k)}$. $w^m_{\hat{x}_s^v(k),\hat{x}_t^v(k)}$ is included in the weight matrix W^m. $D_k = [D_k(1), \ldots, D_k(l), \ldots, D_k(C^t)]$ is the discrepancy vector between $p^s_{\hat{x}_s^v(k)}$ and $p^t_{\hat{x}_t^v(k)}$. D_k is computed as follows:

$$D_k = p^s_{\hat{x}_s^v(k)} T^m - p^t_{\hat{x}_t^v(k)}. \tag{16}$$

Here, each pixel is denoted by a spectral vector; that is, $p^s_{\hat{x}_s^v(k)} = [p^s_{\hat{x}_s^v(k)}(1), \ldots, p^s_{\hat{x}_s^v(k)}(\acute{i}), \ldots, p^s_{\hat{x}_s^v(k)}(C^s)]$, and $p^t_{\hat{x}_t^v(k)} = [p^t_{\hat{x}_t^v(k)}(1), \ldots, p^t_{\hat{x}_t^v(k)}(\acute{j}), \ldots, p^t_{\hat{x}_t^v(k)}(C^t)]$. Ψ is a form of least squares, which has a minimum value.

To minimize Ψ, we find the partial derivatives of Ψ with respect to each element in T^m. The detailed derivatives are shown in Appendix B. All elements in T^m, which minimizes Ψ, can be computed as follows:

$$\begin{bmatrix} T^m(:,1) \\ \vdots \\ T^m(:,\acute{j}) \\ \vdots \\ T^m(:,C^t) \end{bmatrix} = A^{-1}B \tag{17}$$

where

$$A = \begin{bmatrix} E^A & \cdots & 0 & \cdots & 0 \\ \vdots & \ddots & & & \vdots \\ 0 & & E^A & & 0 \\ \vdots & & & \ddots & \vdots \\ 0 & \cdots & 0 & \cdots & E^A \end{bmatrix} \tag{18}$$

$$B = \begin{bmatrix} \left[\sum_{k=1}^{N^x} \left(w^m_{\hat{x}_s^v(k),\hat{x}_t^v(k)} p^t_{\hat{x}_t^v(k)} p^s_{\hat{x}_s^v(k)}{}^T \right)(1,:)\right]^T \\ \vdots \\ \left[\sum_{k=1}^{N^x} \left(w^m_{\hat{x}_s^v(k),\hat{x}_t^v(k)} p^t_{\hat{x}_t^v(k)} p^s_{\hat{x}_s^v(k)}{}^T \right)(\acute{j},:)\right]^T \\ \vdots \\ \left[\sum_{k=1}^{N^x} \left(w^m_{\hat{x}_s^v(k),\hat{x}_t^v(k)} p^t_{\hat{x}_t^v(k)} p^s_{\hat{x}_s^v(k)}{}^T \right)(C^t,:)\right]^T \end{bmatrix} \tag{19}$$

A is a symmetric positive matrix that is always reversible. In (17) and (19), $T^m(:,\acute{j})$ is the $\acute{j}$th column of T^m, and $\sum_{k=1}^{N^x}(w^m_{\hat{x}_s^v(k),\hat{x}_t^v(k)} p^t_{\hat{x}_t^v(k)} p^s_{\hat{x}_s^v(k)}{}^T)(\acute{j},:)$ is the $\acute{j}$th row of $\sum_{k=1}^{N^x}(w^m_{\hat{x}_s^v(k),\hat{x}_t^v(k)} p^t_{\hat{x}_t^v(k)} p^s_{\hat{x}_s^v(k)}{}^T)$. In (18), E^A is a submatrix

$$E^A = \sum_{k=1}^{N^x} w^m_{\hat{x}_s^v(k),\hat{x}_t^v(k)} p^s_{\hat{x}_s^v(k)} p^s_{\hat{x}_s^v(k)}{}^T. \tag{20}$$

Equation (17) is an exact solution given with a finite amount of data, and is with a simple closed form. This leads to really simple steps of AIT. With the category-level relationship between O^t and G^s (as shown in Section III-B3), T^m can be computed using (17). The target-like image I^{st} is then obtained from (13).

D. Assignment-Based Alignment Losses

Unlike existing mainstream methods that perform adversarial learning in feature or output space, we only define two assignment-based alignment losses for features and outputs, respectively, to align the source and target domains. They are also defined based on the solution of the category-level assignment (as shown in Section III-B).

The assignment-based alignment loss L^f_{aa} for features is designed to align the source-domain features f^s and the target-domain features f^t. An average pooling operator (which is similar to that in (10)) is first performed on f^s and f^t so that they can be reshaped into $\mathcal{R}^{C^f \times N^{\text{cla}}}$, where $N^{\text{cla}} = H^{\text{cla}} \times W^{\text{cla}}$. Next, the pointwise distance matrix D^f between f^s and f^t is first computed. Each element in D^f is a square distance. To account for the category-level relationship

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between the source and target domains, $\hat{X}^g$ and W^m as shown in Section III-B3, are also used. Then, L_{aa}^f is computed as follows:

$$L_{\text{aa}}^f = \text{Mean}(D^f \star \hat{X}^g \star W^m) \tag{21}$$

where Mean$(\cdot)$ calculates the average of all elements in a matrix and $\star$ is the element-by-element multiplication.

The assignment-based alignment loss L_{aa}^o for outputs is designed to align the source-domain output O^s and the target-domain output O^t. An average pooling operator is first performed on O^s and O^t in a similar way as the computation of L_{aa}^f. Then, the category-level relationship formed by $\hat{X}^o$ as shown in Section III-B4 is used to compute L_{aa}^o as follows:

$$L_{\text{aa}}^o = \text{Mean}(D^o \star \hat{X}^o \star W^m) \tag{22}$$

where D^o is defined in a similar way as D^f.

L_{aa}^f and L_{aa}^o are used in conjunction with the supervision loss L^s for back-propagation to learn the parameters of the semantic segmentation backbone network. L^s is calculated by the cross-entropy function using the ground-truth G^s and output O^s in the source domain. Concretely, this backpropagation procedure is as follows:

$$\min_{\theta_{\text{net}}} \left(L^s + L_{\text{aa}}^f + L_{\text{aa}}^o\right) \tag{23}$$

where θ_{net} is the parameter set of the semantic segmentation backbone network.

IV. Experiments

In this section, we first introduce the VHR RSI datasets for testing ClA-UDA. Next, the experimental settings are presented. The quantitative and visual results are then shown for each UDA task. Finally, issues of interest, including the ablation study and the impact of the image style transferring based on the category-level assignment and AIT, are discussed.

A. Datasets

To be consistent with existing studies [5], [38], the Potsdam and Vaihingen datasets, which are provided by the International Society for Photogrammetry and Remote Sensing (ISPRS), are employed. Furthermore, RescueNet [46], which is a large-scale natural disaster dataset collected after *Hurricane Michael*, is also employed.

1) Potsdam: The Potsdam dataset provides six semantic classes, including clutter, impervious surfaces, car, tree, low vegetation, and building. It is collected using different imaging modes, and existing UDA studies have selected one or two modes for testing. We opt to maintain consistency with [38] and [5] and select two modes (IR-R-G and R-G-B). Potsdam comprises 38 annotated images with a spatial resolution of 0.05 m/pixel; each image has 6000 × 6000 pixels. To standardize the experiments, we use the official default settings of the Potsdam dataset, i.e., 24 images for training and 14 images for testing, among the 38 images.

2) Vaihingen: The Vaihingen dataset provides six semantic classes, which are the same as those of Potsdam. Vaihingen comprises 33 annotated images with a spatial resolution of 0.09 m/pixel. These images have different sizes and only one imaging mode (i.e., IR-R-G). To make a fair comparison with the accuracy of the Potsdam test set, it is reasonable to divide Vaihingen according to the ratio of training and testing images in Potsdam. As a result, 21 of the 33 images are used for training, and 12 images are used for testing. Specifically, the 12 testing images are the last 12 in the default test set.

3) RescueNet: RescueNet demonstrates the damage to the affected areas caused by *Hurricane Michael* using R-G-B imaging mode. There are 11 semantic classes, including background, debris, water, building_no_damage, building_minor_damage, building_major_damage, building_total_destruction, vehicle, road, tree, pool, and sand. To transfer models from Potsdam to RescueNet, we adjust the category systems of Potsdam and RescueNet. While RescueNet does not contain the low vegetation class, we ignore the low vegetation class in Potsdam when we train the model. For RescueNet, building_no_damage, building_minor_damage, building_major_damage, and building_total_destruction are merged into the building, while debris, water, pool, and sand are merged into the background. In addition to the building and tree classes, the one-to-one correspondence between RescueNet and Potsdam is vehicle ↔ car, background ↔ clutter, and road ↔ impervious surfaces. RescueNet comprises 4494 annotated images with a spatial resolution of 0.015 m/pixel; each image has 4000 × 3000 pixels. Among the 4494 images, 3595 images are for training, 449 are for validation, and 450 are for testing.

B. Experimental Settings

1) Evaluation Metrics: We employ the intersection-over-union (IoU) and F1 scores to evaluate the performance when segmenting each category. The mIoU and mean F1 scores among all existing categories are then computed to show the overall accuracy. Since these metrics are well-known and widely used, we do not present their details due to the page limitation. Their details can be found in [5] and [47].

2) UDA Tasks: Using the Potsdam, Vaihingen, and RescueNet datasets, six UDA tasks are defined as shown in Table I. In Table I, low → high is from a low spatial resolution to a higher resolution. These tasks have three cross-domain variations: 1) geographic locations; 2) imaging modes; and 3) spatial resolutions.

3) Implementation Details: To train ClA-UDA, we use a stochastic gradient descent (SGD) optimizer. The initial learning rate is set to 2.5×10^{-4}, the momentum to 0.9, and the weight decay to 5×10^{-5}. The commonly used learning rate policy is employed, where the initial learning rate is multiplied by $(1 - (\text{iter}/\text{max_iter}))^{\text{power}}$ for each iteration. Here, *power* is set to 0.9. To be consistent with existing studies, such as [5] and [38], we use DeepLabV3 [42] with ResNet101 [48] as the semantic segmentation backbone (baseline model). The space where the category-level assignment is defined in

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TABLE I
UDA TASKS WITH DIFFERENT DOMAIN SHIFTS

UDA tasks		Domain shifts		
		Geographic location	Imaging mode	Spatial resolution
UDA between Potsdam and Vaihingen	Potsdam IR-R-G → Vaihingen IR-R-G	Yes	No	high → low
	Potsdam R-G-B → Vaihingen IR-R-G	Yes	Yes	high → low
	Vaihingen IR-R-G → Potsdam IR-R-G	Yes	No	low → high
	Vaihingen IR-R-G → Potsdam R-G-B	Yes	Yes	low → high
UDA between Potsdam and RescueNet	Potsdam R-G-B → RescueNet R-G-B	Yes	No	low → high
	Potsdam IR-R-G → RescueNet R-G-B	Yes	Yes	low → high

TABLE II
COMPARISON METHODS

Methods	Groups	Modules	Losses	Training stages
MCD [28]	Alignment-based	(1) MCD (2) Dual classifiers	(1) Supervision loss (2) Discrepancy maximization loss (3) Discrepancy minimization loss	One-stop training
ProDA [31]	Self-training-based	(1) Pseudo label denoising (2) Structure learning	(1) Supervision loss (2) Pseudo label training loss (3) Kullback-Leibler loss (4) Loss for regularization	(1) Initial training (2) Pseudo label generation (3) Retraining by pseudo labels
CCDA [5]	UDA for RSIs (Combination of alignment-based and self-training-based strategies)	(1) Entropy-based Patch-wise ranking (2) Adversarial learning (3) Local-to-global feature alignment (4) Semantic-level classifier (5) Entropy-level classifier	(1) Supervision loss (2) Semantic-level loss (3) Entropy-level loss	(1) Training for easy-to-adapt patches (2) Pseudo label generation (3) Training for hard-to-adapt patches
CLAN [13]	Alignment-based (Category-level modeling)	(1) Adversarial learning (2) Dual classifiers	(1) Supervision loss (2) Weight discrepancy loss (3) Category-level adversarial loss	One-stop training
FADA [14]	Alignment-based (Category-level modeling)	(1) Adversarial learning (2) Fine-grained domain discriminator	(1) Supervision loss (2) Discriminator loss (3) Adversarial learning loss	One-stop training
SIM [10]	Alignment-based Self-training-based (Category-level modeling)	(1) Adversarial learning (2) Stuff matching (3) Instance matching	(1) Supervision loss (2) Discriminator loss (3) Adversarial learning loss (4) Instance-level loss (5) Self-training loss	(1) Initial training (2) Pseudo label generation (3) Retraining by pseudo labels
CC-Attention [15]	UDA for RSIs (Category-level modeling)	(1) Adversarial learning (2) Category-certainty attention (3) Class-wise domain predictor	(1) Supervision loss (2) Global adversarial loss (3) Category-level adversarial loss	One-stop training
RC-ADD [16]	UDA for RSIs (Category-level modeling)	(1) Adversarial learning (2) Class clear module (3) Entropy-based regional attention module	(1) Supervision loss (2) Discriminator loss (3) Adversarial learning loss	One-stop training
CaGAN [17]	UDA for RSIs (Category-level modeling)	(1) Transferable attention alignment (2) Per-category selection scheme (3) Adaptive category alignment (4) Adaptive reliably-category selection (5) Adversarial learning	(1) Supervision loss (2) Class-aware domain alignment loss (3) Global alignment loss	(1) Pseudo label generation (2) Alternative and asymptotic training
ClA-UDA(Ours)	UDA for RSIs (Category-level modeling)	(1) Category-level assignment (2) AIT	(1) Supervision loss (2) Alignment-based loss for features (3) Alignment-based loss for outputs	One-stop training

Section III-B3 is set to $\mathcal{R}^{N^c \times N^{\text{cla}}}$, where N^{cla} is $(1/16)H \times (1/16)W$. H and W are the original height and width of the input image patch, respectively.

4) Comparison Methods: For comparison, we select six methods that model the category-level information, and three methods with other characteristics. The detailed information on these methods is shown in Table II. ClA-UDA uses only a one-stop training process and contains fewer modules than CCDA [5], SIM [10], category-certainty attention (CC-Attention) [15], region and category adaptive domain discriminator (RC-ADD) [16], and CaGAN [17].

C. UDA Between Potsdam and Vaihingen

As shown in Table I, we define four UDA tasks between Potsdam and Vaihingen. Both the quantitative and visual results are presented and analyzed.

1) Quantitative Results: Tables III–VI present the quantitative results of ClA-UDA and comparisons with the existing methods. The baseline model (DeepLabV3) obtains the lowest mIoU and mean F1 scores. The baseline model, which does not contain UDA strategies, is trained only by the source-domain samples. This validates that domain shifts indeed exist and UDA methods play a positive role in relieving these domain

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TABLE III
QUANTITATIVE RESULTS OF THE POTSDAM IR-R-G → VAIHINGEN IR-R-G SCENARIO

Methods	mIoU(%)	Mean F1(%)	IoU(%)						F1(%)					
			Impervious surface	Car	Tree	Low vegetation	Building	Clutter	Impervious surface	Car	Tree	Low vegetation	Building	Clutter
Baseline	31.26	43.98	25.52	28.85	52.72	15.48	61.72	3.25	40.66	44.78	69.04	26.80	76.33	6.29
MCD [28]	43.08	57.42	55.20	39.93	61.85	25.27	64.41	11.86	71.13	57.07	76.43	40.34	78.36	21.20
ProDA [31]	44.61	58.64	55.20	41.98	61.01	32.54	68.67	8.24	71.13	59.14	75.78	49.11	81.43	15.22
CCDA [5]	51.93	66.00	**67.78**	46.46	63.26	47.18	71.39	15.52	**80.79**	63.44	77.50	64.11	83.30	26.87
CLAN [13]	43.62	59.11	51.83	35.41	60.98	33.70	60.19	19.58	68.28	52.30	75.76	50.41	75.15	32.75
FADA [14]	44.61	59.88	59.39	44.12	54.39	31.42	61.13	17.22	74.52	61.23	70.46	47.82	75.88	29.37
SIM [10]	46.55	61.63	50.83	40.45	63.38	44.67	64.26	15.72	67.40	57.60	77.59	61.76	78.24	27.17
CC-Attention [15]	49.37	64.12	62.38	44.83	59.51	42.61	69.51	17.38	76.83	61.91	74.61	59.75	82.01	29.62
RC-ADD [16]	50.20	65.50	59.50	48.08	61.31	43.78	65.67	22.89	74.61	64.94	76.01	60.90	79.28	37.26
CaGAN [17]	51.27	66.55	59.40	49.36	63.84	42.41	66.63	25.94	74.53	66.10	77.93	59.56	79.97	41.20
ClA-UDA(Ours)	**55.21**	**69.84**	63.28	**52.91**	**64.11**	**48.03**	**75.13**	**27.80**	77.51	**69.21**	**78.13**	**64.90**	**85.80**	**43.51**

TABLE IV
QUANTITATIVE RESULTS OF THE POTSDAM R-G-B → VAIHINGEN IR-R-G SCENARIO

Methods	mIoU(%)	Mean F1(%)	IoU(%)						F1(%)					
			Impervious surface	Car	Tree	Low vegetation	Building	Clutter	Impervious surface	Car	Tree	Low vegetation	Building	Clutter
Baseline	26.32	37.12	16.07	22.03	51.43	6.03	60.00	2.37	27.69	36.11	67.92	11.38	75.00	4.63
MCD [28]	37.92	51.94	50.09	29.66	54.85	27.44	60.15	5.34	66.74	45.75	70.84	43.06	75.12	10.14
ProDA [31]	38.85	52.70	49.81	36.92	58.48	14.85	50.53	22.53	66.50	53.93	73.80	25.85	67.13	36.77
CCDA [5]	46.27	60.17	51.10	48.53	57.94	31.50	78.01	10.55	67.64	65.35	73.37	47.91	87.65	19.09
CLAN [13]	36.34	50.10	49.00	27.41	53.91	26.81	57.63	3.28	65.77	43.03	70.05	42.28	73.12	6.35
FADA [14]	37.96	52.77	43.93	39.39	46.34	29.71	60.64	7.77	61.05	56.52	63.33	45.80	75.50	14.43
SIM [10]	40.65	54.27	51.80	38.55	59.13	22.49	65.99	5.94	68.24	55.65	74.32	36.72	79.51	11.21
CC-Attention [15]	43.78	58.85	47.02	46.05	58.42	31.84	64.87	14.50	63.96	63.06	73.75	48.30	78.69	25.33
RC-ADD [16]	44.31	58.81	46.39	49.20	58.46	32.75	68.48	10.55	63.38	65.95	73.79	49.34	81.29	19.09
CaGAN [17]	45.35	59.76	55.74	47.30	55.46	31.52	70.44	11.63	71.58	64.22	71.35	47.93	82.66	20.83
ClA-UDA(Ours)	**50.81**	**64.29**	**62.63**	**52.28**	**63.43**	**33.31**	**79.71**	**13.50**	**77.02**	**68.66**	**77.62**	**49.97**	**88.71**	**23.78**

shifts. Our ClA-UDA obtains the highest mIoU and mean F1 scores in all the UDA tasks. The accuracy scores obtained by CCDA rank second. ClA-UDA achieves 3.28%–5.41% higher mIoU scores than CCDA. Among the comparison methods that model category-level information, CaGAN obtains the highest accuracy. ClA-UDA achieves 3.94%–6.81% higher mIoU scores than CaGAN. Although category-level modeling simplifies the training stages of CLAN, FADA, CC-Attention, and RC-ADD, the accuracy problem is still severe. Based on both the quantitative results and the methodological information in Table II, ClA-UDA improves UDA accuracy using fewer modules and only a one-stop training process.

Among the four UDA tasks, the accuracy scores of Potsdam IR-R-G → Vaihingen IR-R-G obtained by all the methods are higher. Given the UDA task information in Table I, we can find the reason. The UDA task of Potsdam IR-R-G → Vaihingen IR-R-G includes only two domain shifts. Furthermore, the spatial resolution variation is from high to low. Compared with the UDA task of Vaihingen IR-R-G → Potsdam IR-R-G, in which the spatial resolution is from low to high, the only difference is in the variation of spatial resolution. This validates that the variation in spatial resolution from low to high is more challenging for UDA methods.

The UDA task of Vaihingen IR-R-G → Potsdam R-G-B challenges UDA methods the most. All the methods obtain lower accuracy. As shown in Table I, Vaihingen IR-R-G → Potsdam R-G-B includes all kinds of domain shifts. Furthermore, the variation in spatial resolution is from low to high. Although these challenges exist, ClA-UDA obtains the highest mIoU and mean F1 scores. More importantly, ClA-UDA achieves the highest IoU and F1 scores in all classes.

2) Visual Results: Figs. 2 and 3 display the visual results of the UDA between Potsdam and Vaihingen. To simultaneously present the overall results and local details, we first show the result of the whole image and then select a representative

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TABLE V
QUANTITATIVE RESULTS OF THE VAIHINGEN IR-R-G → POTSDAM IR-R-G SCENARIO

Methods	mIoU(%)	Mean F1(%)	IoU(%)						F1(%)					
			Impervious surface	Car	Tree	Low vegetation	Building	Clutter	Impervious surface	Car	Tree	Low vegetation	Building	Clutter
Baseline	27.37	40.08	28.83	60.44	14.46	20.59	35.33	4.58	44.76	75.34	25.27	34.15	52.21	8.75
MCD [28]	36.49	50.95	39.59	49.27	41.51	30.19	54.58	3.80	56.72	66.02	58.67	46.37	70.62	7.32
ProDA [31]	37.76	51.29	32.91	55.31	41.02	33.93	62.99	0.43	49.53	71.22	58.17	50.67	77.29	0.85
CCDA [5]	46.82	59.78	60.55	60.48	**50.04**	47.07	62.75	0.28	75.42	75.38	**66.70**	64.01	77.11	0.55
CLAN [13]	37.28	50.67	55.18	58.00	23.97	43.07	43.48	0.00	71.12	73.42	38.67	60.21	60.60	0.00
FADA [14]	40.20	54.09	45.70	52.22	42.25	44.00	57.04	0.0	62.73	68.61	59.41	61.11	72.65	0.00
SIM [10]	40.98	54.45	59.21	55.20	30.10	43.58	56.97	0.81	74.38	71.13	46.28	60.71	72.58	1.61
CC-Attention [15]	43.88	56.78	56.79	65.98	33.36	44.18	62.80	0.15	72.44	79.50	50.03	61.28	77.15	0.30
RC-ADD [16]	46.36	59.12	60.38	65.99	38.58	49.00	64.02	0.21	75.29	79.51	55.68	65.78	78.06	0.41
CaGAN [17]	46.55	59.16	62.51	**67.10**	33.21	49.64	65.75	1.10	76.93	**80.31**	49.86	66.34	79.33	2.17
ClA-UDA(Ours)	**52.23**	**65.80**	**62.74**	65.35	47.74	**54.40**	**72.31**	**10.87**	**77.11**	79.04	64.63	**70.47**	**83.93**	**19.61**

TABLE VI
QUANTITATIVE RESULTS OF THE VAIHINGEN IR-R-G → POTSDAM R-G-B SCENARIO

Methods	mIoU(%)	Mean F1(%)	IoU(%)						F1(%)					
			Impervious surface	Car	Tree	Low vegetation	Building	Clutter	Impervious surface	Car	Tree	Low vegetation	Building	Clutter
Baseline	20.73	30.29	28.32	53.79	1.79	3.47	32.85	4.16	44.14	69.95	3.51	6.70	49.45	7.99
MCD [28]	35.37	49.33	37.21	48.28	39.44	32.98	54.33	0.00	54.24	65.12	56.57	49.61	70.41	0.00
ProDA [31]	36.51	50.63	35.86	43.26	42.61	38.82	57.60	0.92	52.79	60.39	59.76	55.93	73.10	1.83
CCDA [5]	42.84	56.80	47.44	58.71	44.39	37.43	64.31	4.78	64.34	73.98	61.44	54.47	78.28	8.29
CLAN [13]	35.30	49.10	34.02	44.93	37.66	35.30	59.90	0.00	50.77	62.00	54.71	52.18	74.93	0.00
FADA [14]	35.85	49.53	35.13	55.83	40.84	29.47	53.86	0.00	51.99	71.65	58.00	45.52	70.01	0.00
SIM [10]	36.49	50.95	39.59	49.27	41.51	30.19	54.58	3.80	56.72	66.02	58.67	46.37	70.62	7.32
CC-Attention [15]	39.86	53.45	43.41	53.50	39.51	37.42	65.08	0.24	60.53	69.71	56.64	54.46	78.85	0.48
RC-ADD [16]	40.01	54.22	44.08	57.15	41.80	36.79	57.26	2.96	61.19	72.74	58.95	53.79	72.82	5.82
CaGAN [17]	40.74	54.86	48.97	45.77	43.52	38.43	64.75	3.02	65.75	62.80	60.65	55.52	78.60	5.85
ClA-UDA(Ours)	**47.55**	**61.63**	**53.39**	**63.36**	**44.90**	**43.96**	**70.48**	**9.20**	**69.61**	**77.57**	**61.97**	**61.07**	**82.68**	**16.86**

local region to highlight. In addition, to present the differences between different source domains to the same target domain, we use the same target-domain image for visual comparison. In Tables III–VI, CCDA and CaGAN obtain the highest accuracy scores among all the comparison methods. In Figs. 2 and 3, we present the results of CCDA and CaGAN for visual comparison.

In the visual results of the whole image in Fig. 2, the results of Potsdam IR-R-G → Vaihingen IR-R-G are slightly better than those of Potsdam R-G-B → Vaihingen IR-R-G. This validates that the variation in imaging modes poses a challenge to UDA tasks. In Fig. 3, we can obtain the same conclusion. Among the results obtained by different methods, those of the baseline model are the worst. The results of CaGAN and CCDA are similar. When we compare the results of ClA-UDA to those of other methods, we easily find the superiority of ClA-UDA in both the overall results and local details.

D. UDA Between Potsdam and RescueNet

1) Quantitative Results: Tables VII and VIII show the quantitative results of ClA-UDA and the comparisons with existing methods. The baseline model obtains the lowest accuracy. UDA methods improve the accuracy of cross-domain semantic segmentation to varying degrees. As shown in Section IV-A, RescueNet, which contains more images than Potsdam, is very large. Furthermore, RescueNet shows a wide range of damages on buildings and roads that are not included in Potsdam. These issues pose a great challenge to cross-domain semantic segmentation. Accordingly, all the existing methods obtain lower accuracy on the UDA tasks between Potsdam and RescueNet. ClA-UDA obtains the highest mIoU and mean F1 scores on both tasks. ClA-UDA obtains 3.84%–4.89% higher mIoU and mean F1 scores than CCDA which ranks second. For each category, ClA-UDA obtains the highest IoU and F1 scores on three and four out of the five existing

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TABLE VII
QUANTITATIVE RESULTS OF THE POTSDAM R-G-B → RESCUENET R-G-B SCENARIO

Methods	mIoU(%)	Mean F1(%)	IoU(%)					F1(%)				
			Impervious surface	Car	Tree	Building	Clutter	Impervious surface	Car	Tree	Building	Clutter
Baseline	21.46	31.43	26.89	2.47	1.34	23.32	53.27	42.38	4.82	2.64	37.82	69.51
MCD [28]	30.75	43.75	37.44	27.38	0.16	32.11	56.67	54.48	42.98	0.32	48.61	72.34
ProDA [31]	33.35	47.11	28.88	29.52	4.37	49.47	54.52	44.82	45.59	8.37	66.20	70.57
CCDA [5]	36.40	52.10	40.30	31.39	15.83	41.74	52.74	57.45	47.78	27.33	58.89	69.06
CLAN [13]	30.31	42.57	35.28	20.65	0.40	31.30	**63.94**	52.15	34.23	0.79	47.67	**78.00**
FADA [14]	32.48	46.24	12.77	16.84	**24.93**	48.39	59.45	22.65	28.83	39.91	65.22	74.57
SIM [10]	33.62	47.02	39.95	30.79	2.02	41.14	54.18	57.09	45.38	4.03	58.30	70.28
CC-Attention [15]	34.01	48.16	32.90	10.01	22.52	42.77	61.87	49.51	18.20	36.77	59.91	76.44
RC-ADD [16]	35.49	47.96	28.61	24.20	16.14	45.99	62.49	44.49	38.97	16.96	63.01	76.39
CaGAN [17]	36.00	49.07	39.30	31.18	2.20	44.08	63.25	56.43	45.82	4.40	61.19	77.49
ClA-UDA(Ours)	**40.93**	**56.56**	**41.02**	**31.94**	19.54	**51.84**	60.29	**58.18**	**48.42**	32.70	**68.28**	75.22

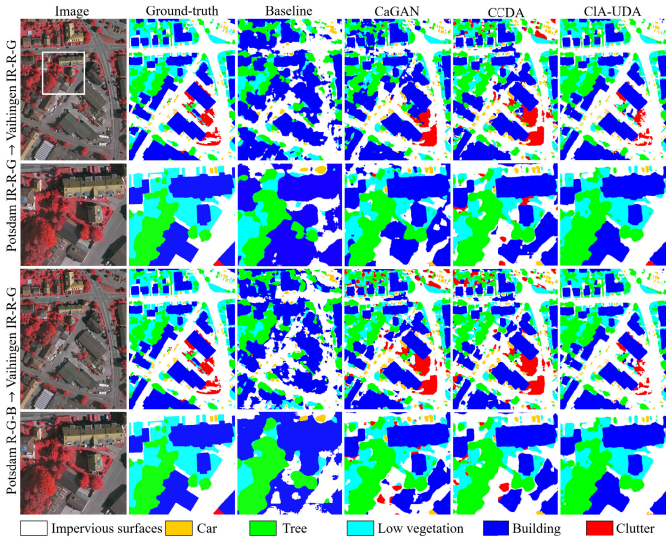


Fig. 2. Visual results of the UDA tasks of Potsdam IR-R-G → Vaihingen IR-R-G and Potsdam R-G-B → Vaihingen IR-R-G.

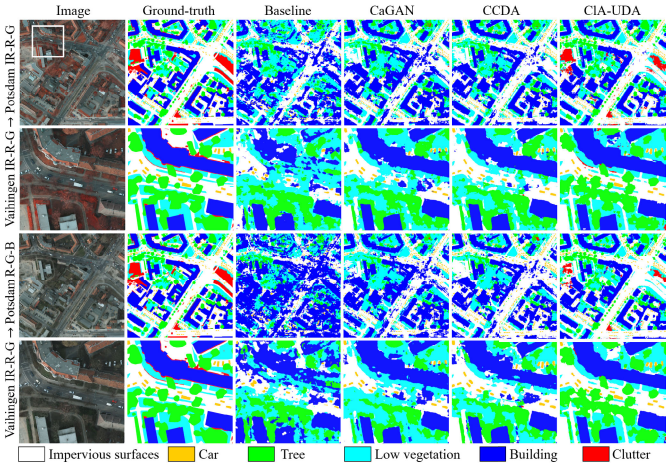


Fig. 3. Visual results of the UDA tasks of Vaihingen IR-R-G → Potsdam IR-R-G and Vaihingen IR-R-G → Potsdam R-G-B.

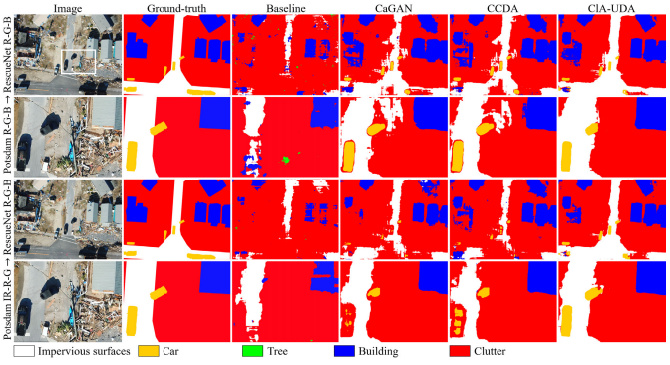


Fig. 4. Visual results of the UDA tasks of Potsdam R-G-B → RescueNet R-G-B and Potsdam IR-R-G → RescueNet R-G-B.

categories in the two tasks, respectively. In particular, damage to impervious surfaces (including roads) and buildings is an important indicator for assessing the impact of disasters. On both tasks, ClA-UDA achieves the highest accuracy for these categories. This further validates the effectiveness of ClA-UDA.

2) Visual Results: Fig. 4 shows the visual results of the UDA between Potsdam and RescueNet. We use the same strategy found in Figs. 2 and 3 to compare different UDA scenarios and methods. RescueNet presents natural disaster damages, with images containing damaged houses and roads covered by debris. This interferes with the model prediction. The comparison in Fig. 4 shows that the buildings and roads recognized by ClA-UDA are the most complete. This demonstrates the application potential of ClA-UDA in disaster emergency response.

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TABLE VIII
QUANTITATIVE RESULTS OF THE POTSDAM IR-R-G → RESCUENET R-G-B SCENARIO

Methods	mIoU(%)	Mean F1(%)	IoU(%)					F1(%)				
			Impervious surface	Car	Tree	Building	Clutter	Impervious surface	Car	Tree	Building	Clutter
Baseline	17.98	29.01	12.77	9.43	40.38	9.70	17.60	22.65	17.24	57.53	17.69	29.93
MCD [28]	24.01	37.84	30.02	16.90	38.74	13.28	21.09	46.18	28.92	55.84	23.44	34.83
ProDA [31]	26.09	39.91	15.83	8.78	41.06	30.86	33.93	27.34	16.14	58.21	47.17	50.67
CCDA [5]	32.63	47.57	33.32	8.56	39.93	34.09	**47.23**	49.99	15.77	57.07	50.84	**64.16**
CLAN [13]	24.91	38.14	19.85	4.59	43.60	26.43	30.10	33.12	8.77	60.73	41.81	46.27
FADA [14]	25.69	39.46	28.83	6.49	38.84	20.15	34.14	44.75	12.19	55.95	33.55	50.90
SIM [10]	29.43	43.59	34.71	3.60	37.52	31.78	39.53	51.53	6.95	54.56	48.24	56.66
CC-Attention [15]	29.49	43.36	24.39	13.16	42.83	33.74	33.35	39.22	18.13	60.85	48.86	49.73
RC-ADD [16]	30.02	45.01	26.34	11.61	43.07	29.91	39.18	41.70	20.80	60.20	46.04	56.30
CaGAN [17]	31.22	45.92	39.78	6.55	39.53	31.33	38.91	56.92	12.30	56.66	47.72	56.02
ClA-UDA(Ours)	**36.47**	**52.46**	**42.63**	**16.47**	**43.89**	**34.33**	45.04	**59.77**	**28.28**	**61.01**	**51.11**	62.11

E. Discussion

1) Ablation Study: There are three novel components in ClA-UDA: category-level assignment (ClA), AIT, and assignment-based alignment losses (AAL). ClA generates a category-level relationship between O^t and G^s [ClR(G)] and a category-level relationship between O^t and O^s [ClR(O)]. AAL consists of two assignment-based alignment losses L_{aa}^f and L_{aa}^o. To validate the impact of each component in a direct way, we remove each component from the complete ClA-UDA. For simplicity, we use ClA-UDA − ClA to represent ClA-UDA without ClA. The other variants of ClA-UDA are denoted in a similar way. The ablation study based on quantitative comparison is shown in Table IX. We select the UDA tasks that contain all three cross-domain shifts between Potsdam and Vaihingen, as shown in Table I. Each component has a positive influence. Specifically, when ClA is removed, the accuracy scores are much lower than those obtained by the other variants. This demonstrates the decisive role of ClA in the performance of ClA-UDA. The accuracy obtained by ClA-UDA − AIT is only higher than those obtained by the baseline model and ClA-UDA − ClA. We can conclude that AIT is another critical component. The accuracy of ClA-UDA − AAL is higher than that obtained by CCDA when we compare the accuracy scores in Tables IV, VI, and IX. ClA-UDA − AAL only uses ClR(G) and AIT, which form a novel image-to-image transferring method. This means that the combination of ClR(G) and AIT also plays a vital role. Fig. 5 shows the ablation study based on visual comparison. The laws of the visual results in Fig. 5 are the same as those of the quantitative results in Table IX, and we can draw the same conclusion.

2) Impact of ClR(G) and AIT: ClR(G) and AIT form a novel image-to-image transferring method (ClR(G) + AIT) that takes category-level relationships into consideration based on ClA. Here, we compare the results of ClR(G) + AIT to those of the commonly used image style transferring methods,

TABLE IX
ABLATION STUDY BASED ON QUANTITATIVE COMPARISON

Methods	Potsdam R-G-B → Vaihingen IR-R-G		Vaihingen IR-R-G → Potsdam R-G-B	
	mIoU(%)	Mean F1(%)	mIoU(%)	Mean F1(%)
Baseline	26.32	37.12	18.80	30.42
ClA-UDA − ClA	39.40	53.45	35.65	50.04
ClA-UDA − AIT	46.11	59.88	40.89	55.75
ClA-UDA − AAL	47.36	60.65	42.95	58.05
ClA-UDA − ClR(G)	41.52	55.33	36.71	51.96
ClA-UDA − ClR(O)	46.75	60.31	41.89	57.95
ClA-UDA − L_{aa}^f	49.93	63.60	45.60	60.11
ClA-UDA − L_{aa}^o	49.27	63.23	44.99	59.72
ClA-UDA	**50.81**	**64.29**	**47.55**	**61.63**

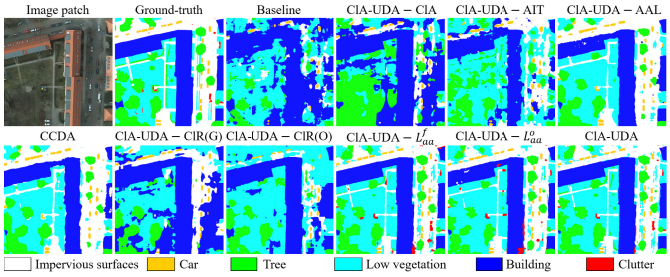


Fig. 5. Ablation study based on visual comparison for the Vaihingen IR-R-G → Potsdam R-G-B scenario.

including CycleGAN [20] and DualGAN [22]. The results are presented in Fig. 6 and Table X.

In Fig. 6, the source-domain images are transformed based on the style of the target-domain images. ClR(G) + AIT

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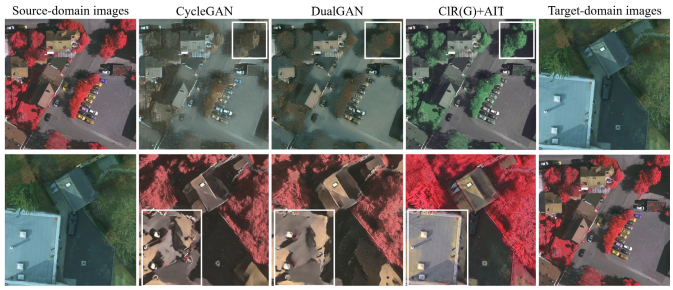


Fig. 6. Visual comparison of different image-to-image transferring methods.

TABLE X
QUANTITATIVE COMPARISON OF DIFFERENT IMAGE-TO-IMAGE TRANSFERRING METHODS

Methods	Potsdam R-G-B → Vaihingen IR-R-G		Vaihingen IR-R-G → Potsdam R-G-B	
	mIoU(%)	Mean F1(%)	mIoU(%)	Mean F1(%)
Baseline + CycleGAN	36.57	51.96	33.89	47.71
Baseline + DualGAN	41.23	56.04	36.13	49.75
Baseline + ClR(G)+AIT	**47.36**	**60.65**	**42.95**	**58.05**

obtains more suitable results compared to CycleGAN and DualGAN. From Vaihingen IR-R-G to Potsdam R-G-B, after the transformation of ClR(G) + AIT, the colors of existing objects are close to those in the target-domain image, especially the trees and low vegetation. From Potsdam R-G-B to Vaihingen IR-R-G, ClR(G) + AIT also obtains better results. To quantitatively compare the performance, we present the accuracy scores obtained by different image-to-image transferring methods in Table X. The combination of the baseline model and ClR(G) + AIT obtains the highest mIoU and mean F1 scores for both UDA tasks.

V. CONCLUSION

This article presents a novel UDA method, ClA-UDA, for the cross-domain semantic segmentation of VHR RSIs. ClA-UDA reveals a crucial mechanism called a category-level assignment, with additional constraints for modeling the category-level relationship between the source and target domains. Subsequently, a novel image-to-image transferring algorithm AIT that minimizes the weighted discrepancy between source-domain and target-domain images is proposed. AIT takes the category-level relationship into consideration and provides an analytical solution to image style transferring. Then, two assignment-based alignment losses are developed using the category-level relationship to align the source and target domains in a concise way. Using only a one-stop training process, ClA-UDA can obtain higher accuracy than state-of-the-art methods.

Three VHR RSI datasets are employed to validate the performance of ClA-UDA. Based on these, six UDA tasks, which have different domain shifts between the source and target domains, are designed. In both the quantitative and visual results, ClA-UDA demonstrates superior performance on all these UDA tasks. To validate the positive influence of each component in ClA-UDA, an ablation study is conducted. The three novel components, including category-level assignment, AIT, and assignment-based alignment losses, improve the accuracy to varying degrees. Specifically, the category-level assignment and AIT play vital roles in improving accuracy.

We believe that a good method should be simple in structure and easy to train. Although ClA-UDA only uses a one-stop training process and has fewer modules, there is still room for improvement. In the future, ClA-UDA will be further simplified while ensuring the accuracy of cross-domain semantic segmentation.

APPENDIX A
RATIONALITY OF $M = N^{\text{cla}} + 1$

The initial solution X^* that minimizes Z in (3) forms N^{cla} initial assigned pairs. An assigned pair $\langle i, j \rangle$ means that the ith source-domain pixel is assigned to the jth target-domain pixel. If we set $M = N^{\text{cla}} + 1$, Additional Constraint B holds, which is equivalent to Proposition 1.

Proposition 1: If we set $M = N^{\text{cla}} + 1$, the N^{cla} assigned pairs have the smallest number of assigned pairs with a cost of M.

To prove Proposition 1, we cannot rely on the procedure used to solve the assignment problem because there are many different methods, all of which can obtain the same solution X^* that minimizes (3). Proving Proposition 1 for each of these methods is a massive project. In each training iteration, the image patches from both domains are randomly selected. When the classifier outputs O^s (or ground-truth G^s) and O^t are transformed into the space $\mathcal{R}^{N^c \times N^{\text{cla}}}$ where the category-level assignment is defined, the shared category set between the source and target domains can be determined using O^s (or G^s) and O^t. Suppose that the shared category set is $S = \{s_1, \ldots, s_k, \ldots, s_m\}$, where $m \leqslant N^c$, and N^c is the number of categories. Furthermore, suppose there are $N^s_{s_k}$ source-domain pixels and $N^t_{s_k}$ target-domain pixels belonging to category s_k. The maximum number $N^{\max}_{s_k}$ of assigned pairs that satisfy Additional Constraint A and (4) for category s_k is

$$N^{\max}_{s_k} = \begin{cases} N^s_{s_k}, & \text{if } N^s_{s_k} < N^t_{s_k} \\ N^t_{s_k}, & \text{otherwise.} \end{cases} \tag{24}$$

Considering all the shared categories, the maximum number $N^{\max}_s$ of assigned pairs that satisfy Additional Constraint A and (4) is

$$N^{\max}_s = \sum_{k=1}^{m} N^{\max}_{s_k}. \tag{25}$$

On the one hand, from (25), we can find that at most $N^{\max}_s$ assigned pixel pairs can be formed, and each pair of assigned pixels belongs to the same category. Additional Constraint B not only makes the assignment comprehensive but also guarantees this result. If Additional Condition B does not hold, at least one of these $N^{\max}_s$ source-domain (or target-domain) pixels will be assigned to a pixel that belongs to a different category, i.e., an assignment error is made. On the other hand, (25) also means that there are at least $N^{\min}_m = N^{\text{cla}} - N^{\max}_s$ assigned pairs with a cost of M. To prove Proposition 1,

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we need to prove that X^*, which minimizes (3), is one of the feasible solutions that contain only $N_m^{\min}$ assigned pairs with a cost of M when M is equal to $N^{\text{cla}}+1$. There are a number of feasible solutions that contain only $N_m^{\min}$ assigned pairs with a cost of M. This is a well-understood permutation problem. Meanwhile, there are also a number of feasible solutions that contain more than $N_m^{\min}$ assigned pairs with a cost of M. Suppose $Z_m^{\min}$ is an objective formulated by a feasible solution that contains only $N_m^{\min}$ assigned pairs with a cost of M, and Z_m is another objective formulated by a feasible solution that contains $N_m^{\min}+N_m^a$ assigned pairs with a cost of M, where $N_m^a \geqslant 1$. $Z_m^{\min}$ and Z_m can be decomposed as follows:

$$\begin{cases} Z_m^{\min} = N_m^{\min} M + \sum_{i'}^{N^{\text{cla}}-N_m^{\min}} c_{i'} \\ Z_m = \left(N_m^{\min}+N_m^a\right) M + \sum_{j'}^{N^{\text{cla}}-N_m^{\min}-N_m^a} c_{j'} \end{cases} \tag{26}$$

where $c_{i'}$ is the cost of the i'th assigned pair. $c_{i'}$ is computed as in (6) and (7), and it is much smaller than M. To guarantee that the optimal solution X^* is one of the feasible solutions that contain only $N_m^{\min}$ assigned pairs with a cost of M, we should ensure that $Z_m^{\min}$ is always smaller than Z_m using $M = N^{\text{cla}}+1$. That is equal to

$$Z_m - Z_m^{\min} = N_m^a M - \left(\sum_{i'}^{N^{\text{cla}}-N_m^{\min}} c_{i'} - \sum_{j'}^{N^{\text{cla}}-N_m^{\min}-N_m^a} c_{j'} \right) > 0. \tag{27}$$

Equation (27) indicates that M should be larger than

$$\begin{aligned} &\left(\sum_{i'}^{N^{\text{cla}}-N_m^{\min}} c_{i'} - \sum_{j'}^{N^{\text{cla}}-N_m^{\min}-N_m^a} c_{j'} \right) \Big/ N_m^a \\ &\quad \leqslant \left(N^{\text{cla}} - N_m^{\min}\right) c^{\max} - \left(N^{\text{cla}} - N_m^{\min} - N_m^a\right) c^{\min} \\ &\quad \leqslant \left(N^{\text{cla}} - N_m^{\min}\right) c^{\max} < N^{\text{cla}} + 1. \end{aligned} \tag{28}$$

As shown in (6) and (7), the maximum value $c^{\max}$ of costs is 1, and the minimum value $c^{\min}$ of costs is 0. Therefore, all the inequalities in Formula (28) hold, and $M = N^{\text{cla}} + 1$ guarantees that Proposition 1 holds. If we use (9) to get the final solution $\hat{X}^*$, $\hat{X}^*$ forms $N_s^{\max}$ assigned pairs, which satisfies Additional Constraint A and Additional Constraint B simultaneously. It is notable that $M = N^{\text{cla}} + 1$ is not extremely large, although M is larger than the other costs in $\mathcal{C}$. For example, when the model is training, image patches with 512×512 pixels are processed. If we use the settings in Section IV-B3, N^{cla} is equal to $(1/16) \times 512 \times (1/16) \times 512 = 1024$. When the model is training, the assignment process does not need any gradient. Computers are capable of handling values in this range. Furthermore, we do not need to calculate the lower bound of M. In each training iteration, the costs in $\mathcal{C}$ are changed. According to Formula (27), to compute the lower bound of M, we need to perform unbalanced assignment programming at least twice in each training iteration. This is not worth the candle.

APPENDIX B
DERIVATIVES OF Ψ IN (15)

To minimize Ψ in (15), we need to find the partial derivative of Ψ with respect to each element in T^m. Taking $T^m(\acute{i}, \acute{j})$ as an example, the partial derivative is

$$\begin{aligned} \frac{\partial \Psi}{\partial T^m(\acute{i}, \acute{j})} &= T^m(1, \acute{j}) \sum_{k=1}^{N^x} w^m_{\hat{x}_s^v(k), \hat{x}_t^v(k)} p^s_{\hat{x}_s^v(k)}(1) p^s_{\hat{x}_s^v(k)}(\acute{i}) + \cdots \\ &\quad + T^m(\acute{i}, \acute{j}) \sum_{k=1}^{N^x} w^m_{\hat{x}_s^v(k), \hat{x}_t^v(k)} p^s_{\hat{x}_s^v(k)}(\acute{i})^2 + \cdots \\ &\quad + T^m(C^s, \acute{j}) \sum_{k=1}^{N^x} w^m_{\hat{x}_s^v(k), \hat{x}_t^v(k)} p^s_{\hat{x}_s^v(k)} \\ &\quad \times (C^s) p^s_{\hat{x}_s^v(k)}(\acute{i}) \\ &\quad - \sum_{k=1}^{N^x} w^m_{\hat{x}_s^v(k), \hat{x}_t^v(k)} p^s_{\hat{x}_s^v(k)}(\acute{i}) p^t_{\hat{x}_t^v(k)}(\acute{j}). \end{aligned} \tag{29}$$

When we set the above formula equal to zero and move the terms that are unrelated to T^m to the right side, we get the following equation:

$$\begin{aligned} &T^m(1, \acute{j}) \sum_{k=1}^{N^x} w^m_{\hat{x}_s^v(k), \hat{x}_t^v(k)} p^s_{\hat{x}_s^v(k)}(1) p^s_{\hat{x}_s^v(k)}(\acute{i}) + \cdots \\ &\quad + T^m(\acute{i}, \acute{j}) \sum_{k=1}^{N^x} w^m_{\hat{x}_s^v(k), \hat{x}_t^v(k)} p^s_{\hat{x}_s^v(k)}(\acute{i})^2 + \cdots \\ &\quad + T^m(C^s, \acute{j}) \sum_{k=1}^{N^x} w^m_{\hat{x}_s^v(k), \hat{x}_t^v(k)} p^s_{\hat{x}_s^v(k)}(C^s) p^s_{\hat{x}_s^v(k)}(\acute{i}) \\ &= \sum_{k=1}^{N^x} w^m_{\hat{x}_s^v(k), \hat{x}_t^v(k)} p^s_{\hat{x}_s^v(k)}(\acute{i}) p^t_{\hat{x}_t^v(k)}(\acute{j}). \end{aligned} \tag{30}$$

Considering all elements in T^m, we get the following equation in matrix form based on (30):

$$A \begin{bmatrix} T^m(:, 1) \\ \vdots \\ T^m(:, \acute{j}) \\ \vdots \\ T^m(:, C^t) \end{bmatrix} = B \tag{31}$$

where A is shown in (18), and B is shown in (19).

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